

Nikhilesh Vijay Chirutala

Python LLM Engineer | Software Engineer

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SUMMARY

Python Software Engineer with 3 years of experience building and maintaining production applications, backend services, APIs, and cloud-based systems, with hands-on experience developing LLM and RAG-powered applications. Experienced in Python, FastAPI, REST APIs, microservices, AWS, SQL, MongoDB, vector search, LLM integration, prompt engineering, and AI-powered workflows. Built a RAG-based financial chatbot and an MCP integration enabling an AI bot to retrieve application data and trigger supported operations. Experienced in authentication and authorization, API integrations, workflow automation, production observability, CI/CD, troubleshooting, and reliable deployment of distributed services.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript (ES6+), TypeScript

LLM & Generative AI: LLM Integration, RAG, Prompt Engineering, AI-Powered Applications, AI-Assisted Workflows, MCP Integration, Structured API Integration

RAG & Retrieval: Retrieval-Augmented Generation, Vector Search, Contextual Retrieval, Knowledge Retrieval, Prompt Engineering

Backend & Microservices: Python, FastAPI, REST APIs, API Development, Microservices, Backend Automation, WebSockets

AI & Tool Integration: MCP, API Integration, External Service Integration, Tool-Based AI Workflows, Authentication, Authorization, ACL, RBAC

Data & Databases: MongoDB, MySQL, PostgreSQL, Neon DB, SQL, Data Processing, Data Validation

Cloud & DevOps: AWS, Amazon S3, AWS Lambda, GitHub Actions, CI/CD, Vercel, Linux

Observability & Reliability: Datadog, Grafana, Sentry, Monitoring, Troubleshooting, Error Handling, Production Support

Development Tools: Git, GitHub, Postman, Unit Testing, API Testing, Debugging

WORK EXPERIENCE

Software Engineer • **CA One Tech Cloud Inc.**

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog

- Develop and maintain production-grade Python services and APIs for a high-volume cloud contact-center platform, supporting automated processing, data workflows, storage operations, and production systems.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, implementing automated processing, feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, processing limits, retries, response validation, and result persistence.
- Developed automated validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, inconsistent storage states, and error scenarios across production environments.
- Supported production deployment and rollout of Python services across multiple clusters, including configuration, validation, monitoring, troubleshooting, deployment verification, and rollback support.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage processes.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry to monitor service health, track errors, diagnose processing failures, and support production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing root-cause analysis and implementing fixes to maintain reliable and resilient processing workflows.

Software Engineer • **Mantra Technologies**

Oct 2023 – Mar 2026 | Hyderabad, India

- Delivered production features and backend services across multiple applications, translating business requirements into APIs, application workflows, integrations, and end-to-end technical solutions.
- Developed Python and FastAPI services and REST APIs supporting business processes, external integrations, data flows, and AI-enabled application capabilities.
- Implemented authentication, authorization, ACL-based access control, external service integrations, validation, and workflow logic while troubleshooting integration and application issues.
- Supported production releases through testing, debugging, root-cause analysis, deployment validation, and defect resolution.

Key Projects:

AI Bull — AI-Powered Trading & Investment Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts into a unified application.
- Built Python and FastAPI backend services supporting portfolio aggregation, investment analysis, screening, backtesting, and AI-powered application workflows.

- Implemented a Retrieval-Augmented Generation (RAG) chatbot that combines portfolio and financial information with LLM capabilities to provide contextual, portfolio-aware responses to user queries.
- Implemented contextual retrieval workflows to provide relevant portfolio and financial information to the LLM for generating grounded responses.
- Applied prompt engineering and application-level validation to structure AI responses around user portfolio context and supported financial use cases.
- Developed stock and mutual fund screening workflows with configurable filtering and data-driven analysis capabilities.
- Designed and implemented a backtesting workflow for evaluating trading strategies against historical market data and analyzing strategy performance.

CRM.io — AI-Driven CRM & Automation Platform

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM

- Developed Python and FastAPI backend services and REST APIs for an AI-driven CRM, implementing business workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built a real-time CRM dialer integrating Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts provided for relevant calls.
- Implemented task cadence workflows that allow users to create sequences of tasks as a single workflow and progress through scheduled follow-ups, with Firebase Cloud Messaging used for notifications.
- Built document management workflows using Amazon S3, including browser-to-S3 uploads and automated association of documents with CRM records.
- Implemented a CRM-wide soft-delete and retention workflow using AWS Lambda to automate permanent removal of records after the retention period.
- Built account-level reporting dashboards and developed an MCP integration exposing CRM capabilities as tools, enabling an AI bot to retrieve CRM information and trigger supported operations.
- Applied ACL-based authorization to control access to CRM capabilities and application operations exposed through backend services and integrations.

PERSONAL PROJECTS

[JS Studio](#) — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

[AnyScrum](#) — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur